

MSNet: Multimodal Self-attention Network for Depression Detection via Fusion of Eye Tracking and EEG

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Abstract

Amidst the multimodality of biomedical data and the escalating prevalence of depression, multimodal biosignal fusion stands out as a crucial trend in depression research. Our study introduces a novel Multimodal Self-Attention Network (MSNet), which integrates the multi-head attention mechanism simultaneously to eye tracking and EEG data, enhancing depression detection through long-range dependencies modeling ability. MSNet automatically extracts attention distribution from eye tracking data, utilizing multi-head attention mechanism with location coding to capture attention bias globally. For EEG data, MSNet incorporates residual structures into graph attention network to transmit neighbor aggregation features through multi-head attention mechanism, facilitating dynamic weighted distribution between nodes and neighbors to enhance feature consistency. With a classification accuracy of 93.62%, MSNet showed a significant enhancement compared to using singular eye tracking or EEG data, showcasing the benefits of combining both modalities in depression detection and the effectiveness of the self-attention mechanism in processing physiological signals.

CCS Concepts

• Computing methodologies → Neural networks.

Keywords

Multimodal, Signal processing, Depression detection, Eye tracking, Transformer



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1 Introduction

Depression is a pervasive mental disorder, characterized by low mood and poor concentration, ultimately impacting an individual's ability to communicate. According to the World Health Organization, over 280 million people globally suffer from depression, making it a significant public health concern [Organization 2001]. The presence of a depressed mood in youth poses a risk for the eventual development of depressive disorders later in life. Within the domain of youth mental health, approximately 40% undergo a resurgence of depression within a 3–5-year span [Hankin 2006]. Scale analysis and doctor-patient communication are the primary methods for diagnosing depression. However, subjective biases in scale assessments and patient reluctance to cooperate can lead to insufficient detection accuracy [LIU and ZHOU 2015], it is urgent and essential to find accurate and objective biomarkers for depression detection [Jamal et al. 2023].

Recently, multimodal biosignal fusion has emerged as a promising approach for detecting mental disorders by synthesizing data from various modes [Muhammad et al. 2021], offering diverse perspectives on common medical phenomena to address inference challenges. The biosignal modes include eye tracking [Armstrong and Olatunji 2012; Skaramagkas et al. 2021], electrocardiogram (EOG) [Pan et al. 2021], and electroencephalography (EEG) [Yasin et al. 2021, 2023], the relationships between these modes can be

categorized into three types: complementarity, redundancy, and collaboration [Stahlschmidt et al. 2022]. Integrating EEG and eye tracking is classified under collaboration since the combined information offers a holistic view compared to individual data sources. Zhu et al. [Zhu et al. 2022] successfully identified depression by combining EEG features, yielding an accuracy rate of 85.83%. Upon integrating the deep belief network (DBN) with eye tracking data, the accuracy rate significantly improved to 89.54%, a noteworthy enhancement of 3.71%.

Feature extraction is crucial in multimodal biosignal fusion [Lahat et al. 2015]. Neural networks excel in this regard, showing superior performance in depression detection across various medical data types [Kim et al. 2022; Merlin Praveena et al. 2022]. Convolutional neural networks (CNNs) are effective for continuous data analysis, particularly in image tasks. Lin et al. [Lin et al. 2021] integrated emotional images and eye tracking to detect depression with 90.06% accuracy using CNNs. However, for irregular domains like brain networks, CNNs face challenges in spatial convolution operations [Craik et al. 2019]. Graph neural networks (GNNs) redefine operations in the Fourier domain, enabling better representation learning with EEG data [Graña and Morais-Quilez 2023], surpassing CNNs in handling diverse electrode spatial relationships effectively.

In our study, we utilized Psychopy to construct a multimodal acquisition platform that synchronizes eye tracking and EEG signals, enhancing the robustness of multimodal data integration. Our research delves into the influence of fusion stages on classification efficacy, exploring early, intermediate, and late fusion in multimodal data fusion. Our work can be succinctly summarized as follows:

- 1) We propose the Multimodal Self-Attention Network (MSNet), leveraging a multi-head attention mechanism to simultaneously process eye tracking and EEG signals. This approach maximizes the model's capacity for robust long-distance dependence modeling, facilitating the effective fusion of features and complementary information from different modes. The classification accuracy of MSNet achieves 93.62%, demonstrating the potential of utilizing multimodal biological signal fusion for depression detection.

- 2) Regarding the eye tracking data, we established an Eye Vision Transformer (EVT) module within MSNet. This module transforms one-dimensional eye tracking data into two-dimensional images, utilizing a combination of the multi-head attention mechanism and location coding to automatically extract attention distributions during experiments. This method captures attention bias globally and locally, enabling the detection of depression group from varied perspectives.

- 3) Concerning the EEG data, we introduced the Residual graph attention (RGAT) module in MSNet. This module integrates a residual structure into the graph attention network, transmitting neighbor aggregation features obtained through the multi-head attention mechanism to nodes as residual information. This enhances feature consistency, ensuring stable signal transmission while preserving the intrinsic signal contributions of the nodes.

2 Materials and methods

2.1 Subjects and Data Acquisition

Within our research, we enrolled 47 adolescents from Xi'an Cheliang High School, consisting of 21 individuals with depression risk

(Mean age = 15.4, SD age = 0.72) and 26 healthy control participants (Mean age = 15.8, SD age = 0.84). The exclusion criteria involved cases with less than 80% eye tracking data or incomplete EEG experiments, leading to the exclusion of 5 subjects. The study adhered to the principles outlined in the Declaration of Helsinki and received approval from the Ethics Committee of the Third Affiliated Hospital of Anhui Medical University (Ref.: 2020-020-01), located in Hefei, China. Written informed consent was obtained from all participants and their parents.

The DSM-V assessment scale [Bragazzi and Del Puente 2014] was utilized in this study, with individuals scoring above 32 on the DSM-V scale being classified as the depression risk group. The study incorporated an attentional paradigm involving sudden pedestrian appearances and vehicle collisions in 20 trials. This paradigm demonstrated significant differences between depression risk groups and healthy controls by quantifying individuals' attentional patterns in driving-oriented tasks [Wang et al. 2023]. The EyeCatch Bar by Dongguan Dongquan Technology Co., Ltd., with 1.25 degrees precision and 41 Hz sampling rate, was used for eye tracking. A 64-channel HydroCel Geodesic Sensor Net recorded EEG at 250 Hz, keeping impedance below 50k and calibrated to Cz. Preprocessing in MATLAB (version 2017b) and EEGLAB isolated brainwave activity and removed interference [Delorme and Makeig 2004], applying filters, baseline correction, and ICA for artifact removal to enhance EEG signal quality.

2.2 The Architecture of Multimodal Self-Attention Network

The architecture of our proposed MSNet is depicted in Figure 1. On one hand, for eye tracking data, MSNet incorporates the Eye Vision Transformer (EVT) module to transform the two-dimensional coordinate data into heat images through visual mapping. By integrating the multi-head attention mechanism with location coding, MSNet autonomously extracts attention distributions from eye tracking data throughout the experiment, capturing attention biases comprehensively. This includes identifying variations in visual attention areas and gaze duration during the experiment. On the other hand, for EEG data, MSNet employs the Residual graph attention (RGAT) module to incorporate the residual structure into the graph attention network. The neighbor aggregation features derived from the multi-head attention mechanism are transmitted to the nodes as residual information, enabling dynamic weighted distribution among nodes and their neighbors while enhancing feature transmission consistency and preserving the intrinsic signal contributions of the nodes.

2.2.1 Eye Vision Transformer Module. As illustrated in Figure 1, EVT segmented the image displayed by the experimental paradigm into individual blocks, with each block representing a distinct region. Through visual mapping, we generated heat maps from the eye tracking coordinates captured within 30 seconds across a total of 20 trials, these images were then employed as inputs to the model [Dosovitskiy et al. 2020]. To sum up, we reshaped the image into a flattened sequence of 2D patches, conducted $E \in \mathbb{R}^{H \times W \times C}$ to $E \in \mathbb{R}^{(P^2 \cdot C)}$ conversion. In this process, (H, W) represents the original image resolution, C signifies the number of channels, (P, P) denotes the patch resolution, and N corresponds to the number of

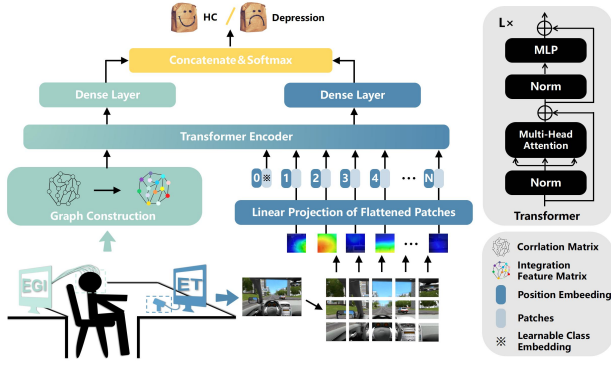


Figure 1: The architecture of our proposed MSNet. Eye tracking data and EEG data are initially processed independently and subsequently fed into the Transformer encoder for self-attention feature extraction. The yellow layer illustrates the stage of data fusion.

patches computed by $N = HW/P^2$. Clearly, this process ensures that each slice represents a distinct trial. The Transformer maintains a consistent latent vector size D across its layers, yielding patch embeddings as output.

$$z'_f = \text{MSA}(\text{LN}(z_{f-1})) + z_{f-1}, \quad f = 1 \cdots F \quad (1)$$

$$z_f = \text{MLP}(\text{LN}(z'_f)) + z'_f, \quad f = 1 \cdots F \quad (2)$$

To encode positional details, we add a learnable class embedding at the beginning of the sequence of embedded patches. Employing standard learnable 1D position embeddings, we construct a sequence of embedding vectors for the encoder input. The resulting state at the output of the Transformer encoder functions as the image representation.

The Transformer encoder is structured with stacked layers that incorporate multi-head self-attention (MSA) [Wang et al. 2022] and MLP blocks in alternating fashion, where a GELU non-linearity operates across two layers within the MLP block, and represent the outputs of MSA and MLP, respectively, with denoting the classification heads. Before each block, layernorm (LN) is applied, followed by residual connections after each block. Subsequently, the eye tracking data is processed by transformer encoder to produce a dense layer. This layer is then concatenated with the vector derived from behavioral characteristics for classification.

2.2.2 Residual Graph Attention Module. In the Residual graph attention module, EEG channels act as nodes in a graph where edges are established through Pearson correlation analysis and threshold-based binarization, creating an adjacency matrix. This matrix is fed into a transformer encoder with a two-layer graph attention network. The first layer combines residual framework and multi-head attention with LeakyReLU activation for neighbor feature integration. The second layer performs classification using single-head attention and softmax activation. Dropout and regularization are included to improve network robustness.

The framework of the graph construction layer is illustrated in Figure 2. The input of the layer comprises a set of node features,

$\mathbf{X} = \{x_1, x_2, x_3, \dots, x_N\}$, $x_i \in \mathbb{R}^F$, where N represents the number of nodes, F signifies the feature dimension of each node, and denotes the output feature dimension. To significantly boost the expressive capacity in transitioning input features to higher-level features, it is imperative to integrate a learnable linear transformation. Initially, a shared linear transformation using the weight matrix $\mathbf{W} \in \mathbb{R}^{F \times F'}$, is applied individually to every node. Subsequently, the self-attention mechanism $a : \mathbb{R}^F \times \mathbb{R}^{F'} \rightarrow \mathbb{R}$ is computed to determine the attention coefficients:

$$e_{ij} = a(\mathbf{W}_{x_i}, \mathbf{W}_{x_j}) \quad (3)$$

This emphasizes the importance of node j 's feature concerning node i . We incorporate the graph structure into the mechanism using masked attention, computing e_{ij} exclusively for node $j \in V_i$, where V_i represents node i 's neighborhood in the graph. To facilitate easy comparison of coefficients across various nodes, we employ the softmax function to normalize them across all j options.

$$a_{ij} = \text{softmax}_j(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k \in V_i} \exp(e_{ik})} \quad (4)$$

In our experiments, the attention mechanism is realized through a single-layer feedforward neural network, incorporating the weight vector $\alpha \in \mathbb{R}^{2F'}$, and leveraging the LeakyReLU nonlinear activation function with a negative input slope of $\alpha = 0.2$. Upon complete expansion, the coefficients computed by the attention mechanism can be represented as:

$$a_{ij} = \frac{\exp(\text{LeakyReLU}(\alpha^T [\mathbf{W}_{x_i} \parallel \mathbf{W}_{x_j}]))}{\sum_{k \in V_i} \exp(\text{LeakyReLU}(\alpha^T [\mathbf{W}_{x_i} \parallel \mathbf{W}_{x_k}]))} \quad (5)$$

Where T represents transposition and $\parallel$ is the concatenation operation.

Upon acquiring the normalized attention coefficients, utilize them to compute the linear combination of the respective features. Integrated with the multi-head attention mechanism, this calculation forms a residual variable that is fused with the node's original signal, ultimately serving as the final output feature for each node.

$$x'_i = x_i + \sigma \left(\frac{1}{K} \sum_{k=1}^K \sum_{j \in V_i} a_{ij}^k \mathbf{W}_{x_j}^k \right) \quad (6)$$

Where K represents the heads, $\mathbf{W}^k$ is the weight matrix for head k , σ is the sigmoid activation function, and x'_i represents the output of layer.

2.3 Fusion stage and Training

In our experiment, the fusion stage was divided into early, intermediate, and late fusion [Boulahia et al. 2021]. Early fusion encountered challenges due to varying sample rates across modalities, using FCNs for classification performance assessment with manually extracted multimodal features. The manually extracted features comprise three eye tracking features and four EEG features. The former consists of AOI sample percentage, AOI fixation time, and fixation distribution [Over et al. 2006], while the latter includes power spectral density, activity, mobility, and complexity. Conversely, deep learning architectures are better suited for intermediate fusion, which involves the fusion concatenation of

Table 1: Comparison of different fusion stages and their performances.

Fusion stage	Hyper parameters	Selection range	Accuracy
Early fusion	Network node	32, 16, 2	78.48%
		64, 16, 2	80.71%
Intermediate fusion ¹	Network node	64, 2	91.49%
		128, 2	89.36%
Intermediate fusion ²	Network node	64, 2	91.52%
		128, 2	93.62%
Late fusion	Weighted method	Average Weighted	91.54%
		Entropy Weighted	85.11%

feature vectors to enhance performance. The late fusion involved processing eye tracking and EEG patterns independently to create sub-models, utilizing average and entropy weighting methods. In the case of entropy weighting, a weight ratio of 3:4 was chosen based on the extraction dimensions from the manually extracted features.

The baseline comparison model comprised two components: one for eye tracking data and the other for EEG analysis. The eye tracking comparison model encompassed ResNet [He et al. 2016], GoogleNet [Szegedy et al. 2015], and SENet [Hu et al. 2018]. The EEG comparison models consisted of GCN [Kipf and Welling 2016], Transformer [Vaswani et al. 2017], FBCNet [Mane et al. 2021], and EEG Conformer [Song et al. 2022]. To ensure a comprehensive performance assessment, we employed various evaluation metrics, including accuracy, F1 score, sensitivity, and specificity. The model training process entailed the utilization of the Cross-Entropy loss function and the Adam optimization algorithm, executed on an NVIDIA RTX 3090 GPU. Within the PyTorch framework in Python 3.8, we conducted the training process, setting the learning rate to 0.001. In our experiment, we employed a five-fold cross-validation method to ensure robust evaluation. The final accuracy was obtained by averaging the results from the five cross-validation folds. The default setting assigns to 3, representing the three channels of the image, and are configured as 1080 and 1920, the size of patches is designated as 120, and is computed as 144. The number of heads in Transformer encoder is set to 4, and the threshold for adjacency matrix is set to 0.7. We included the process of parameter selection in the supplementary material.

3 Results

3.1 The Classification Results based on different fusion stages

Our study analyzed the impact of early, intermediate, and late fusion in multimodal data analysis, with the results summarized in Table 1. Intermediate fusion showed superior performance in processing EEG and eye tracking data, effectively handling the diversity of multimodal datasets. Early fusion involved manual extraction and fusion of data, achieving a classification accuracy of 80.71%. Late fusion using average weighting achieved a notable accuracy of 91.54%. In examining the impact of the residual network in intermediate fusion within the MSNet framework, results showed that excluding

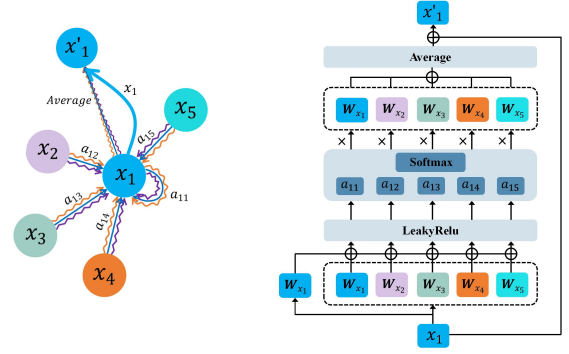


Figure 2: The workflow of the proposed Residual graph attention module. The left image illustrates the interplay among diverse node operations in real space, while the right image delineates the calculation process of the proposed algorithm.

the residual network yielded an accuracy of 91.49%. By incorporating the residual structure with a larger intermediate layer, the accuracy improved to 93.62%. The enhancements in accuracy, sensitivity, specificity, and F1-score were noteworthy, demonstrating the superior performance of the MSNet network. Both intermediate and late fusion methods surpassed early fusion in classification performance.

Table 2: Performance comparison of different models.

Model	Accuracy	Sensitivity	Precision	F1-score
ResNet	80.85%	84.62%	76.92%	0.806
GoogLeNet	76.60%	69.23%	84.62%	0.762
SENet	80.71%	80.777%	80.95%	0.808
EVT	87.23%	88.46%	86.54%	0.875
GCN	82.98%	76.92%	88.4%	0.818
Transformer	89.36%	90.48%	88.32%	0.899
FBCNet	83.04%	84.54%	81.48%	0.830
EEG Conformer	85.1%	84.75%	86.54%	0.855
RGAT	91.49%	92.31%	90.36%	0.914
MSNet (Ours)	93.62%	96.15%	90.48%	0.932

3.2 The Classification Results based on MSNet

In our exploration of fusion classification tasks within deep learning framework, we initially performed classification evaluations on independent eye tracking modal, with the corresponding results summarized in the first four rows of Table 2. Notably, our proposed EVT module exhibited outstanding performance in this classification task, achieving the highest classification accuracy of 87.23%. When compared to models like ResNet, GoogLeNet, and SENet, the EVT module demonstrated notable performance in critical classification metrics, including specificity (88.46%), sensitivity (86.54%), and F1 score (0.875), underscoring its efficacy in processing eye tracking data. This highlights the potential for achieving superior classification results through the lightweight eye tracking platform.

Subsequently, we conducted the evaluation on EEG modal, as detailed in rows 5 to 9 of Table 2. The analysis revealed that all baseline comparison models achieved classification accuracies exceeding 80%, with the RGAT module standing out by attaining a classification accuracy of 91.49%, coupled with a sensitivity of 92.31%, a specificity of 90.36%, and an F1-score of 0.914. These four fundamental classification metrics placed the RGAT module at the forefront among all individual EEG modal classification models.

Finally, we delved into the fusion strategy involving the EVT module and RGAT module, leveraging the intermediate fusion method established in prior studies to enhance classification efficiency. Following the fusion, the model achieved a classification accuracy of 93.62%, marking a notable improvement of 6.39% and 2.13% compared to the individual eye tracking and EEG mode, respectively. To effectively showcase our experimental results, we compared MSNet with other tasks that employ multi-modal methods for disease detection or mood classification [33–37], as outlined in Table 3. It's essential to note the distinct datasets utilized in each of these works.

Table 3: Performance comparison of different models.

References	Classifier	Multimodal	Fusion
Ge et al.	SKLDA	EEG, EYE	84.54%
Wang et al.	Transformer	EEG, EYE	90.02%
Zhu et al.	Bayesian Rule	EEG, EYE	91.12%
Koelstra et al.	Naive Bayes	EEG, facial expressions	86.70%
Guo et al.	SVM	EEG, EYE, Eye Image	79.63%
Ours	MSNet	EYE, EEG	93.62%

4 Discussion and conclusion

The objective of our study is to establish a multimodal signal acquisition platform utilizing eye tracking and EEG to investigate the visual attention mechanism and cognitive brain activity among individuals with depression risk. Introducing the novel Multimodal Self-Attention Network (MSNet), which leverages multi-head attention mechanisms at its core, enables comprehensive capture of complementary information from eye tracking and EEG data with robust long-range modeling capabilities. The achieved classification accuracy surpasses baseline models, reaching 93.62%, underscoring the potential of multimodal biological signal fusion in depression

detection. Besides, the fusion stage emerges as crucial, with intermediate fusion demonstrating superior feature extraction capabilities, particularly in capturing modal feature interactions.

The fusion of multimodal data exhibits enhanced feature extraction abilities compared to single modal data, significantly boosting classification performance and mitigating the impact of sample imbalance on results, notably in accuracy and F1-score. The accuracy of the MSNet framework reached 93.62%, marking a substantial increase of 6.39% and 2.13% compared to eye tracking and EEG modes, aligning with previous research findings on depression detection [Ge et al. 2022; Wang et al. 2021]. Notably, sensitivity in depression detection significantly improved through multimodal fusion, with intermediate and late fusion achieving sensitivities of 96.15% and 92.26%, respectively, surpassing other baseline algorithms and showcasing potential in preventing misdiagnosis during diagnosis [Guo et al. 2019; Koelstra and Patras 2013; Zhu et al. 2022].

In dealing with eye tracking data, the EVT module within MSNet maps one-dimensional eye tracking signals to two-dimensional images, automatically extracting attention patterns using a multi-head attention mechanism with location coding. The EVT module achieves an accuracy of 87.23%, with an F1-score of 0.875, indicating its capability to substitute traditional manual eye tracking feature extraction and emphasizing the efficacy of eye tracking as a lightweight physiological signal.

For EEG data processing, the RGAT module in MSNet integrates residual structures into the graph attention network, transmitting neighbor aggregation features obtained through the multi-head attention mechanism as residual information to realize dynamic weighted distribution between nodes and neighbors. This enhancement significantly boosts model classification performance, with the final accuracy and F1-score increasing by 2.13% and 0.018, respectively, post introduction of the residual structure, reaching an impressive 93.62% and 0.932. The successful application of residual structures in this study, aligning with prior research [Lu and Triesch 2019; Srivastava et al. 2021], highlights its broad utility in enhancing multi-modal data analysis performance.

In this study, the algorithm's performance is constrained by the high cost of acquiring multimodal data and the limited dataset size. As a result, there is an anticipation to streamline the data acquisition equipment and modes in the future to mitigate acquisition expenses. The aim is for this research to serve as a screening tool for depression, with intentions to progressively extend its application to middle schools.

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